

Manufacturing Downtime Analysis - Database Design

Executive Summary

This document provides comprehensive technical documentation for the data model architecture underlying the Manufacturing Downtime Analysis Power BI dashboard. The design follows dimensional modeling best practices using a star schema approach to enable efficient querying, filtering, and analytical capabilities.

1. Data Architecture Overview

1.1 Data Source

Primary Source: Manufacturing_Line_Productivity.xlsx

Format: Microsoft Excel Workbook

Location: Local file system

Refresh Method: Manual on-demand refresh

1.2 Data Model Pattern

Schema Type: Star Schema

Fact Tables: 1 (Line downtime LU)

Dimension Tables: 3 (Products LU, Downtime factors LU, Line productivity LU)

Relationship Type: One-to-Many from dimensions to fact

1.3 Design Principles

- **Simplicity:** Easy to understand and maintain
- **Performance:** Optimized for fast query response
- **Flexibility:** Supports ad-hoc analysis and filtering
- **Scalability:** Can accommodate growth in data volume
- **Maintainability:** Clear naming conventions and documentation

2. Table Definitions

2.1 Fact Table: Line downtime LU

Purpose: Contains granular downtime event records at the batch-operator-factor level

Grain: One row per downtime occurrence

Row Count: Variable (hundreds to thousands)

Refresh Frequency: On-demand

Schema

Column Name	Data Type	Description	Sample Values
Batch	Text	Batch identifier for production run	"B001", "B002", "B003"
ProductID	Whole Number	Foreign key to Products LU	1, 2, 3, 4, 5, 6
OperatorID	Whole Number	Foreign key to Line productivity LU	1, 2, 3, 4
FactorID	Whole Number	Foreign key to Downtime factors LU	1, 2, 3, 4, 5, 6, 7, 8
Minutes	Whole Number	Duration of downtime event in minutes	15, 30, 45, 60
StartDT	Date/Time	Start timestamp of downtime event	"2024-08-29 08:00:00"
EndDT	Date/Time	End timestamp of downtime event	"2024-08-29 08:30:00"

Relationships

- ProductID → Products LU[ProductID] (Many-to-One)
- OperatorID → Line productivity LU[OperatorID] (Many-to-One)
- FactorID → Downtime factors LU[FactorID] (Many-to-One)

Data Validation Rules

- Minutes > 0
- EndDT > StartDT
- All foreign keys must exist in respective dimension tables
- No null values allowed in key columns

Indexes (Recommended for large datasets)

- Primary: Batch + ProductID + OperatorID + FactorID
- Foreign Keys: ProductID, OperatorID, FactorID

2.2 Dimension Table: Products LU

Purpose: Lookup table for product information

Grain: One row per unique product

Row Count: 6

Type: Slowly Changing Dimension (Type 1)

Schema

Column Name	Data Type	Description	Sample Values
ProductID	Whole Number	Primary key, unique identifier	1, 2, 3, 4, 5, 6
Product	Text	Product code/name	"CO-2L", "CO-600", "DC-600", "LE-600", "OR-600", "RB-600"

Relationships

- ProductID ← Line downtime LU[ProductID] (One-to-Many)

Data Quality Rules

- ProductID must be unique
- Product names must be non-null and unique
- Consistent naming convention (XX-NNN format)

2.3 Dimension Table: Downtime factors LU

Purpose: Lookup table for downtime root cause categories

Grain: One row per unique downtime factor

Row Count: 8

Type: Slowly Changing Dimension (Type 1)

Schema

Column Name	Data Type	Description	Sample Values
FactorID	Whole Number	Primary key, unique identifier	1, 2, 3, 4, 5, 6, 7, 8
Factor	Text	Short code for downtime factor	"4", "6", "Factor", "8", etc.
Description	Text	Detailed description of factor (optional)	"Machine breakdown", "Material shortage"

Relationships

- FactorID ← Line downtime LU[FactorID] (One-to-Many)

Data Quality Rules

- FactorID must be unique
- Factor codes must be non-null
- Consistent categorization across time periods

2.4 Dimension Table: Line productivity LU

Purpose: Lookup table for operator and production line metadata

Grain: One row per operator-product-date combination

Row Count: Variable (hundreds)

Type: Slowly Changing Dimension (Type 2 with date tracking)

Schema

Column Name	Data Type	Description	Sample Values
OperatorID	Whole Number	Primary key, unique identifier	1, 2, 3, 4
Operator	Text	Operator name	"Charlie", "Dee", "Dennis", "Mac"
ProductID	Whole Number	Foreign key to Products LU	1, 2, 3, 4, 5, 6
Date	Date	Production date	"2024-08-29", "2024-08-30", "2024-08-31"
RuntimeDurationMinutes	Whole Number	Total scheduled runtime	480, 500, 520
MinimumRuntime	Whole Number	Minimum expected runtime	400, 420, 450

Calculated Columns

StartDT (Date/Time)

Purpose: Derived start datetime from Date

Formula: `Date + TIME(8, 0, 0)` (assumes 8 AM start)

EndDT (Date/Time)

Purpose: Derived end datetime from Date and RuntimeDurationMinutes

Formula: `StartDT + (RuntimeDurationMinutes / 1440)` (convert minutes to days)

Relationships

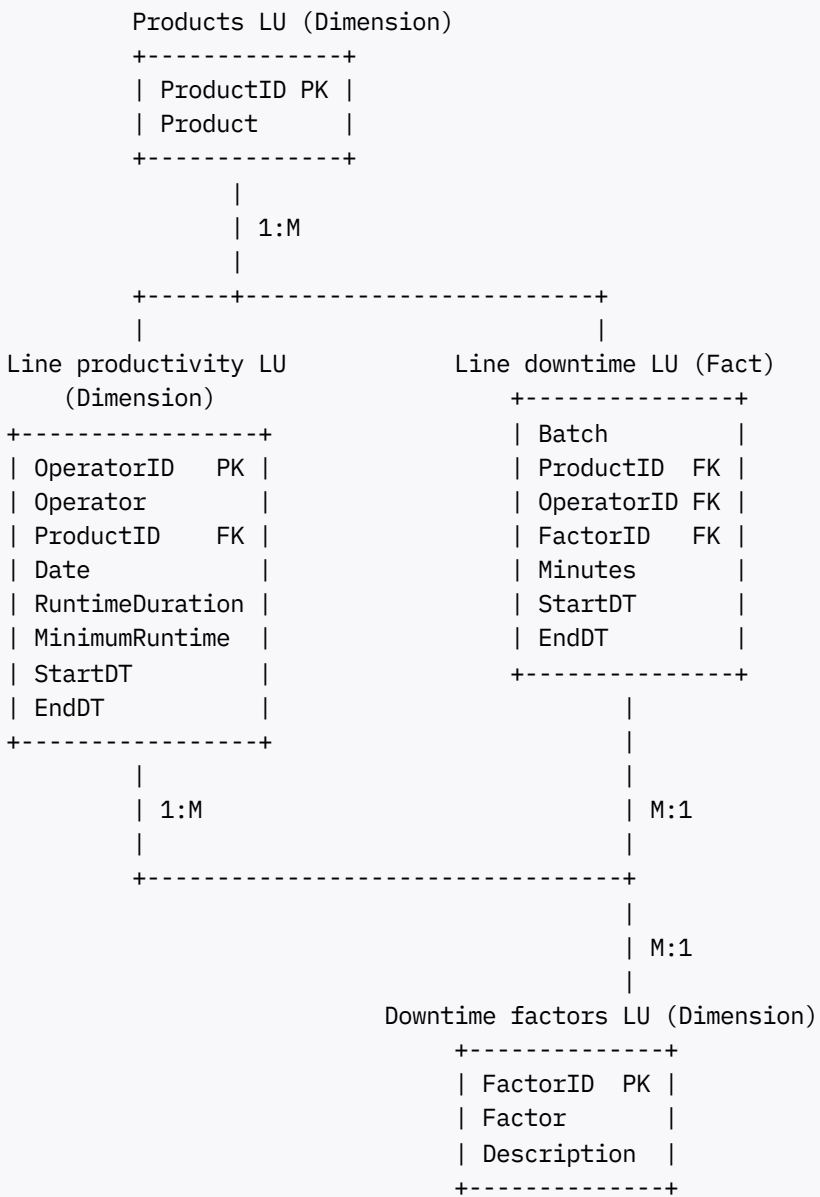
- `OperatorID` ← Line downtime LU[OperatorID] (One-to-Many)
- `ProductID` → Products LU[ProductID] (Many-to-One)

Data Quality Rules

- OperatorID must be unique per date-product combination
- RuntimeDurationMinutes > MinimumRuntime
- Dates must be valid business days
- No future dates allowed

3. Relationship Model

3.1 Star Schema Diagram



3.2 Relationship Properties

From Table	From Column	To Table	To Column	Cardinality	Cross-Filter Direction	Active
Line downtime LU	ProductID	Products LU	ProductID	Many-to-One	Single (to Products)	Yes
Line downtime LU	OperatorID	Line productivity LU	OperatorID	Many-to-One	Single (to Line productivity)	Yes
Line downtime LU	FactorID	Downtime factors LU	FactorID	Many-to-One	Single (to Downtime factors)	Yes
Line productivity LU	ProductID	Products LU	ProductID	Many-to-One	Single (to Products)	Yes

3.3 Relationship Validation

- **Referential Integrity:** All foreign keys in fact table must have matching primary keys in dimension tables
- **No Circular Relationships:** Model is acyclic (no circular dependencies)
- **Bidirectional Filtering:** Not used to avoid ambiguous filter propagation
- **Inactive Relationships:** None (all relationships are active)

4. Calculated Columns

4.1 In Line productivity LU

StartDT

Purpose: Convert Date to DateTime with assumed start time

Formula:

```
StartDT = 'Line productivity LU'[Date] + TIME(8, 0, 0)
```

Rationale: Enable time-based calculations and datetime filtering

EndDT

Purpose: Calculate end datetime based on runtime duration

Formula:

```
EndDT = 'Line productivity LU'[StartDT] + ('Line productivity LU'[RuntimeDurationMinutes]
```

Rationale: Support duration calculations and timespan analysis

5. DAX Measures

5.1 Core Measures

Net Runtime Minutes

Purpose: Calculate net productive runtime (Runtime - Downtime)

Formula:

```
Net Runtime Minutes = [Total Runtime Minutes] - [Total Downtime Minutes]
```

Usage: Primary KPI card, Matrix visual

Format: Whole Number, 0 decimal places

Total Runtime Minutes

Purpose: Sum of all scheduled runtime across selected context

Formula:

```
Total Runtime Minutes = SUM('Line productivity LU'[RuntimeDurationMinutes])
```

Usage: Internal calculation, Matrix visual

Format: Whole Number, 0 decimal places

Total Downtime Minutes

Purpose: Sum of all downtime events in selected context

Formula:

```
Total Downtime Minutes = SUM('Line downtime LU'[Minutes])
```

Usage: KPI card, Matrix visual, Trend chart

Format: Whole Number, 0 decimal places

Variance vs Min

Purpose: Calculate difference between actual runtime and minimum expected

Formula:

```
Variance vs Min =  
VAR ActualRuntime = [Total Runtime Minutes]  
VAR MinRuntime = SUM('Line productivity LU'[MinimumRuntime])  
RETURN  
ActualRuntime - MinRuntime
```

Usage: KPI card, Matrix visual
Format: Whole Number, 0 decimal places

Downtime%

Purpose: Calculate downtime as percentage of total runtime
Formula:

```
Downtime% =  
DIVIDE(  
    [Total Downtime Minutes],  
    [Total Runtime Minutes],  
    0  
)
```

Usage: KPI card, Matrix visual, Conditional formatting
Format: Percentage, 2 decimal places

6. Data Dictionary

Complete Field Reference

Table	Field	Type	Description	Constraints
Line downtime LU	Batch	Text	Batch production identifier	Not null
Line downtime LU	ProductID	Integer	Product foreign key	FK → Products LU
Line downtime LU	OperatorID	Integer	Operator foreign key	FK → Line productivity LU
Line downtime LU	FactorID	Integer	Downtime factor foreign key	FK → Downtime factors LU
Line downtime LU	Minutes	Integer	Downtime duration	> 0
Line downtime LU	StartDT	DateTime	Downtime start timestamp	Not null
Line downtime LU	EndDT	DateTime	Downtime end timestamp	> StartDT
Products LU	ProductID	Integer	Unique product identifier	PK, Unique
Products LU	Product	Text	Product code/name	Not null, Unique
Downtime factors LU	FactorID	Integer	Unique factor identifier	PK, Unique
Downtime factors LU	Factor	Text	Factor short code	Not null

Table	Field	Type	Description	Constraints
Downtime factors LU	Description	Text	Detailed factor description	Optional
Line productivity LU	OperatorID	Integer	Unique operator identifier	PK
Line productivity LU	Operator	Text	Operator name	Not null
Line productivity LU	ProductID	Integer	Product foreign key	FK → Products LU
Line productivity LU	Date	Date	Production date	Not null
Line productivity LU	RuntimeDurationMinutes	Integer	Scheduled runtime	> 0
Line productivity LU	MinimumRuntime	Integer	Minimum expected runtime	> 0, < RuntimeDuration
Line productivity LU	StartDT	DateTime	Calculated start time	Calculated column
Line productivity LU	EndDT	DateTime	Calculated end time	Calculated column

7. Data Loading & ETL

7.1 Extract Process

Source: Manufacturing_Line_Productivity.xlsx

Method: Power Query (M)

Steps:

1. Connect to Excel workbook
2. Load worksheets as separate queries
3. Promote headers
4. Set data types
5. Remove empty rows

7.2 Transform Process

Products LU

```
let
    Source = Excel.Workbook(File.Contents("Manufacturing_Line_Productivity.xlsx")),
    Products_Sheet = Source{[Name="Products"]}[Data],
    PromotedHeaders = Table.PromoteHeaders(Products_Sheet),
```

```

    ChangedType = Table.TransformColumnTypes(PromotedHeaders,{
        {"ProductID", Int64.Type},
        {"Product", type text}
    })
in
    ChangedType

```

Downtime factors LU

```

let
    Source = Excel.Workbook(File.Contents("Manufacturing_Line_Productivity.xlsx")),
    Factors_Sheet = Source{Name="Downtime Factors"}[Data],
    PromotedHeaders = Table.PromoteHeaders(Factors_Sheet),
    ChangedType = Table.TransformColumnTypes(PromotedHeaders,{
        {"FactorID", Int64.Type},
        {"Factor", type text}
    })
in
    ChangedType

```

Line productivity LU

```

let
    Source = Excel.Workbook(File.Contents("Manufacturing_Line_Productivity.xlsx")),
    Productivity_Sheet = Source{Name="Line Productivity"}[Data],
    PromotedHeaders = Table.PromoteHeaders(Productivity_Sheet),
    ChangedType = Table.TransformColumnTypes(PromotedHeaders,{
        {"OperatorID", Int64.Type},
        {"Operator", type text},
        {"ProductID", Int64.Type},
        {"Date", type date},
        {"RuntimeDurationMinutes", Int64.Type},
        {"MinimumRuntime", Int64.Type}
    })
in
    ChangedType

```

Line downtime LU

```

let
    Source = Excel.Workbook(File.Contents("Manufacturing_Line_Productivity.xlsx")),
    Downtime_Sheet = Source{Name="Line Downtime"}[Data],
    PromotedHeaders = Table.PromoteHeaders(Downtime_Sheet),
    ChangedType = Table.TransformColumnTypes(PromotedHeaders,{
        {"Batch", type text},
        {"ProductID", Int64.Type},
        {"OperatorID", Int64.Type},
        {"FactorID", Int64.Type},
        {"Minutes", Int64.Type},
        {"StartDT", type datetime},
        {"EndDT", type datetime}
    })

```

```

    }),
    FilteredRows = Table.SelectRows(ChangedType, each [Minutes] > 0)
in
    FilteredRows

```

7.3 Load Process

Destination: Power BI Data Model

Mode: Import (In-Memory)

Compression: VertiPaq columnar compression

Refresh Method: Manual on-demand

8. Data Quality Framework

8.1 Validation Rules

Rule	Table	Field	Validation	Action on Failure
Positive Minutes	Line downtime LU	Minutes	> 0	Filter out row
Valid Date Range	Line productivity LU	Date	Between 2024-01-01 and Today	Alert user
FK Integrity	Line downtime LU	ProductID	Exists in Products LU	Error, halt load
FK Integrity	Line downtime LU	OperatorID	Exists in Line productivity LU	Error, halt load
FK Integrity	Line downtime LU	FactorID	Exists in Downtime factors LU	Error, halt load
Runtime Logic	Line productivity LU	RuntimeDuration	> MinimumRuntime	Alert user
DateTime Logic	Line downtime LU	EndDT	> StartDT	Filter out row
Uniqueness	Products LU	ProductID	Unique values	Error, halt load

8.2 Data Profiling

Recommended Checks Before Load:

- Row count per table matches expectations
- No null values in key columns
- All foreign keys have matching dimension records
- Date ranges are within expected boundaries
- Numeric fields contain valid numbers (no text)
- Duplicate primary keys are absent

9. Performance Optimization

9.1 Model Optimizations

- **VertiPaq Compression:** Enabled by default in Import mode
- **Cardinality:** Proper Many-to-One relationships for optimal query plans
- **Column Data Types:** Use smallest appropriate data type (e.g., Int16 vs Int64)
- **Remove Unused Columns:** Delete columns not needed for analysis
- **Disable Auto Date/Time:** Prevent automatic date hierarchy tables

9.2 DAX Optimizations

- **DIVIDE() Function:** Safe division with default zero handling
- **SUM() vs SUMX():** Use SUM for aggregated columns, SUMX only when needed
- **Variable Usage:** Store intermediate calculations in variables
- **Filter Context:** Leverage filter propagation instead of explicit filters where possible

9.3 Query Optimizations

- **Limit Visual Rows:** Use Top N filtering to reduce data transfer
- **Aggregations:** Pre-aggregate data at grain appropriate for analysis
- **Incremental Refresh:** Consider for large historical datasets (future enhancement)

10. Security & Governance

10.1 Data Classification

Sensitivity Level: Internal Use Only

Contains PII: Yes (Operator names)

Compliance Requirements: Internal HR policies

10.2 Access Control

Method: Power BI Service Row-Level Security (future implementation)

Roles:

- **Admin:** Full access to all data
- **Manager:** Access to all operators and products
- **Supervisor:** Access to assigned operators only

RLS DAX Expression (Example for Supervisor role):

```
[Operator] = USERPRINCIPALNAME()
```

10.3 Data Retention

- Historical Data:** Retain 2 years in active model
- Archive Strategy:** Export to data warehouse for long-term storage
- Backup Frequency:** Weekly Power BI file backup

11. Change Management

11.1 Version Control

- Current Version:** 1.0
- Change Log:**
 - v1.0 (Oct 2025): Initial release with Overview page

- Future Versions:**
 - v1.1 (Planned): Add Batch Detail page
 - v1.2 (Planned): Implement RLS
 - v2.0 (Planned): Integrate with MES system

11.2 Schema Changes

- Process:**
 - Document proposed change
 - Impact analysis on existing measures and visuals
 - Test in development copy
 - Update documentation
 - Deploy to production
 - Notify stakeholders

12. Troubleshooting Guide

Common Issues

Issue	Symptom	Resolution
Relationship not working	Filters don't propagate	Check cardinality and cross-filter direction
Blank values in visual	No data displayed	Verify foreign keys match dimension tables
Incorrect totals	Aggregation doesn't match expectations	Review DAX measure context and filter propagation
Slow refresh	Data load takes >5 minutes	Optimize Power Query steps, reduce data volume

Issue	Symptom	Resolution
Circular dependency error	Cannot create relationship	Redesign model to remove circular path

Conclusion

This database design document provides a comprehensive blueprint for the Manufacturing Downtime Analysis data model. The star schema architecture ensures optimal query performance, clear relationships, and flexibility for future enhancements. By adhering to dimensional modeling best practices and maintaining rigorous data quality standards, the model serves as a robust foundation for actionable production insights.